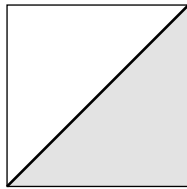


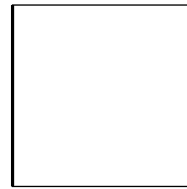
Use whole squares inside the gray shape. Cover all squares with some gray area. A touch at an edge or corner adds no square.



whole

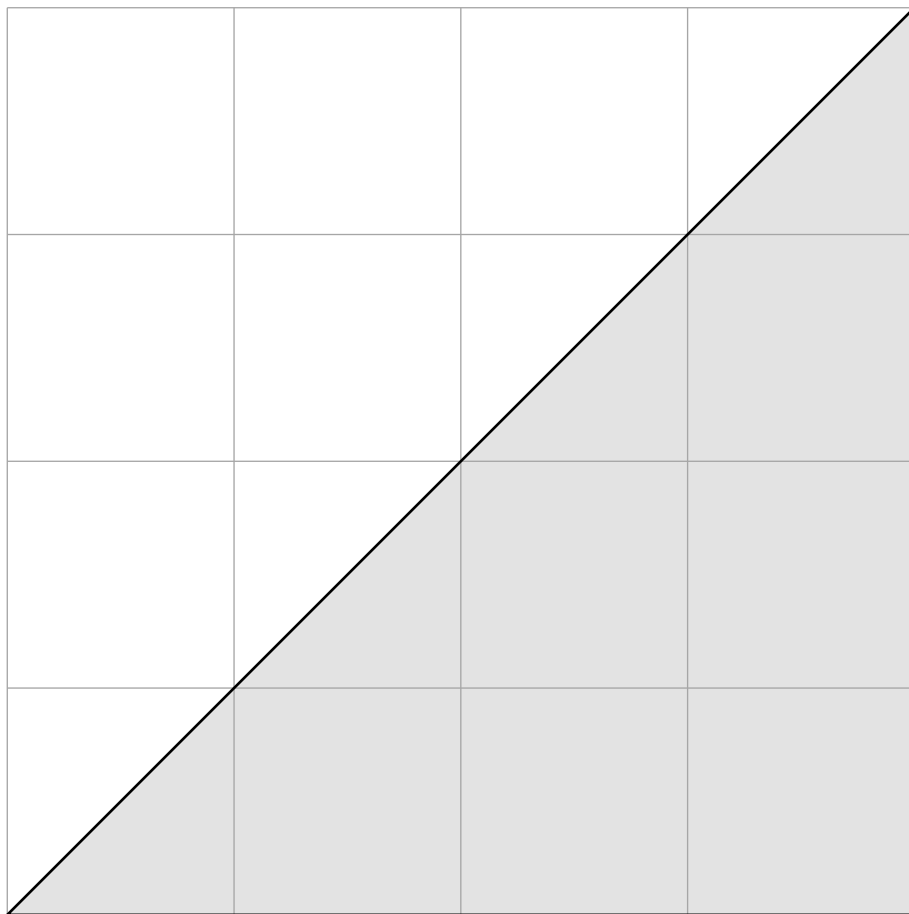


partial



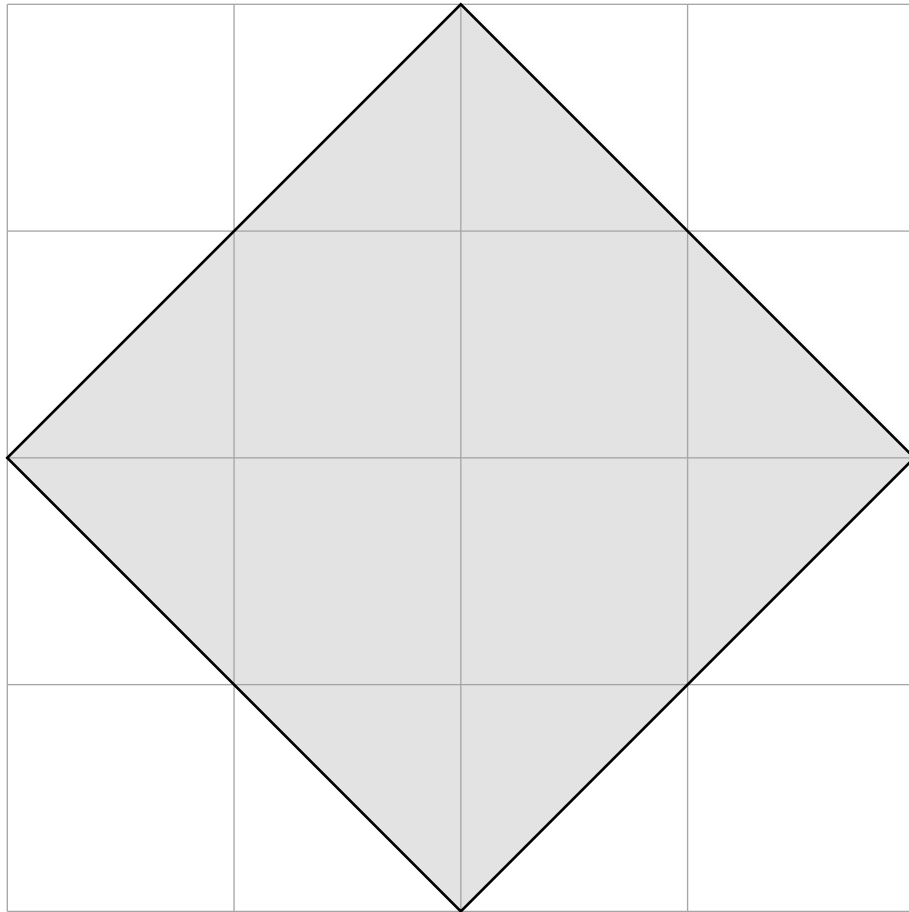
outside

Problem 1: Find an inside cover and an outside cover for the triangle. Use as many whole inside squares and as few outside-cover squares as possible.



Inside: _____ big squares Cover: _____ big squares

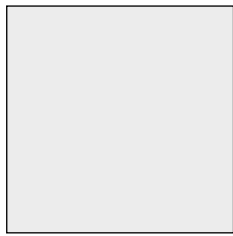
Problem 2: Trap the diamond's area between an inside cover and an outside cover. Can you be certain its area is more than 3 big squares? Less than 13?



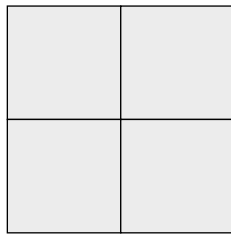
Inside: _____ big-square units

Cover: _____ big-square units

One grid square is 1 big-square unit.

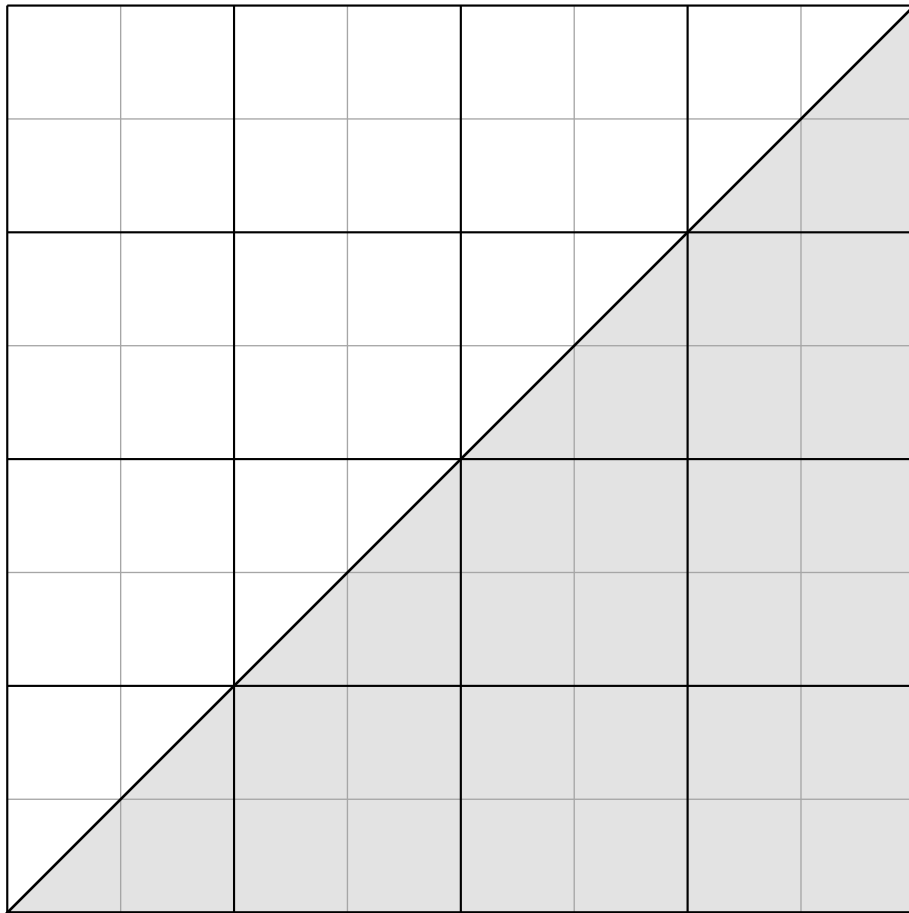


=



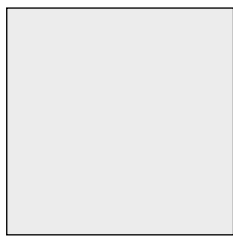
1 big square
= 4 small squares

Problem 3: Use the smaller squares to trap the triangle's area more tightly. State both bounds in big-square units.

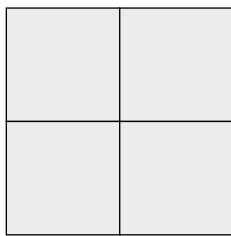


Inside: _____ big-square units

Cover: _____ big-square units

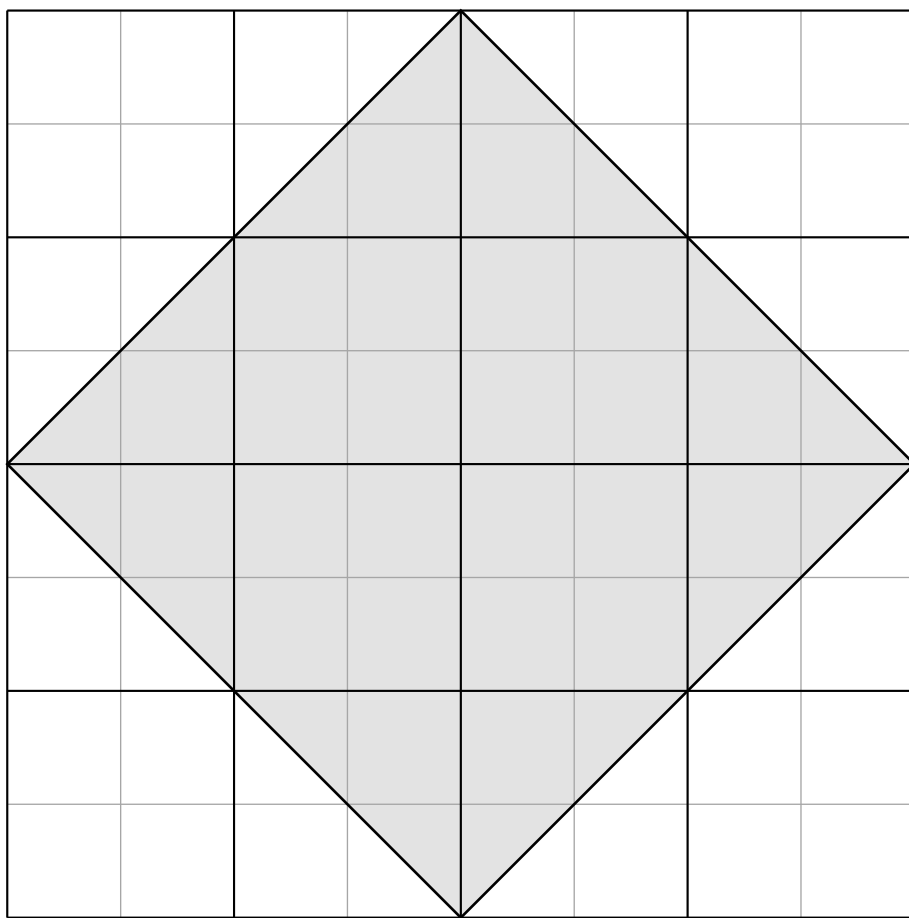


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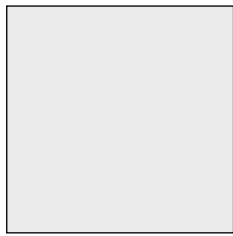
1 big square
= 4 small squares

Problem 4: Find new bounds for the diamond. Compare the area left uncertain on the big grid and on the small grid.

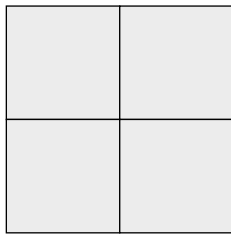


Inside: _____ big-square units

Cover: _____ big-square units

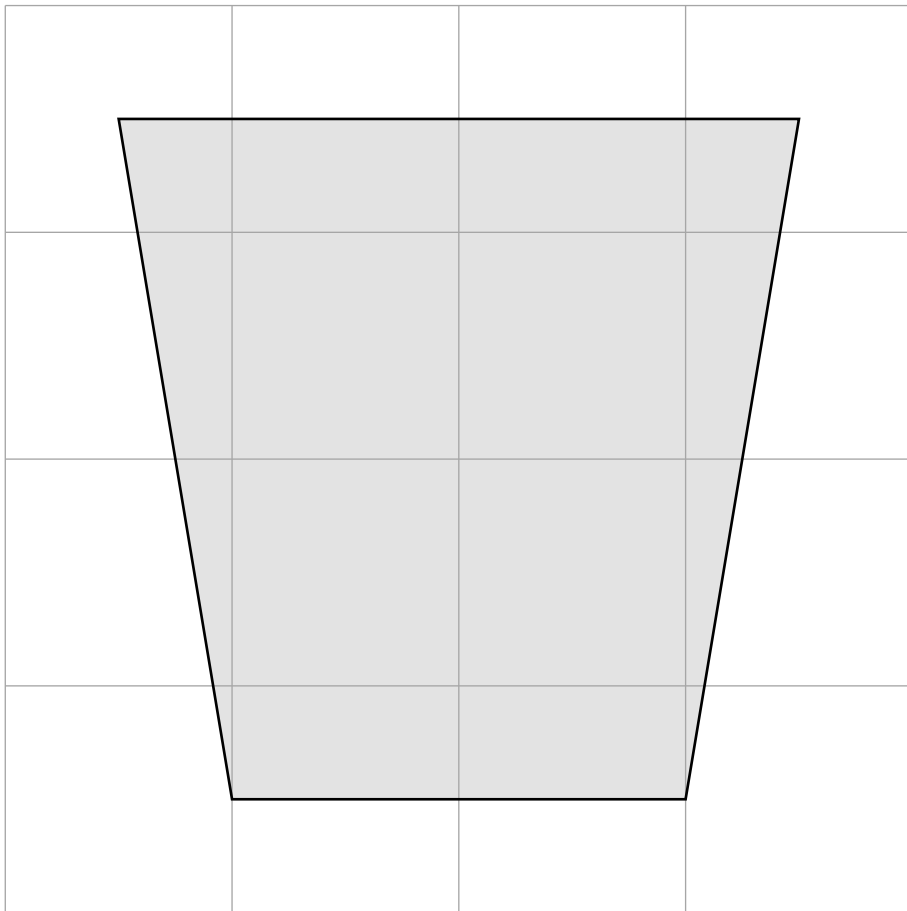


=



1 big square
= 4 small squares

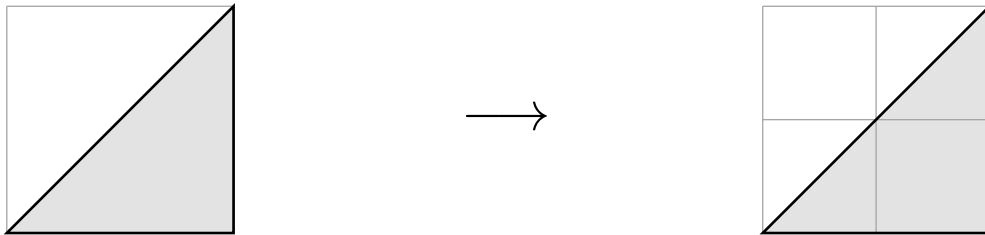
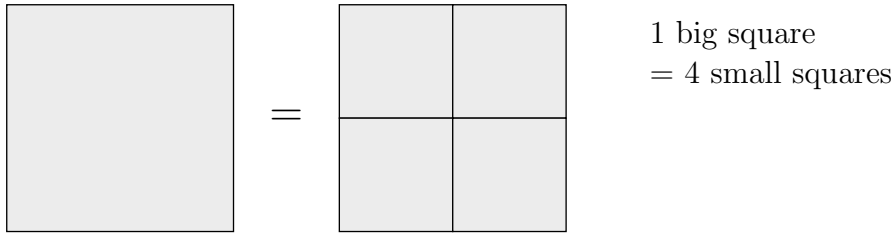
Problem 5: Split exactly three big squares into four small squares each. Choose the three that most reduce the gap between your inside and outside bounds.



Inside: _____ big-square units

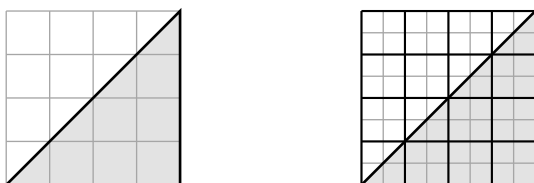
Cover: _____ big-square units

Problem 6: A big square is split into four small squares. Could the total inside area go down? Could the outside-cover area go up? Explain why your answers work for any of these shapes.



Problem 7: The original triangle on page 1 fits with a second copy into its 4-by-4 square. Find its exact area. Explain how that area fits between both pairs of bounds you found.

Smaller pictures of the original triangle; not to scale.



Exact area: _____
big-square units